

Andrey Zakharov

Python Backend Developer

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Python Backend Developer with 3+ years of hands-on project-based development experience. Builds API-driven applications, data-processing services and integrations with external systems, including simulation engines, AI pipelines, media workflows and messaging platforms. Experienced in taking backend-heavy projects from architecture and MVP implementation to testing, packaging and deployment.

Experience

Independent Project Development: 2023 – Present.

COMPASS AI – Python Backend Engineer

2026 – Present

Tech: Python, FastAPI, Pydantic, Pandas, NumPy, REST APIs, Plane API, CSV/JSON/Parquet, ONNX Runtime

- Designed and implemented FastAPI backend services for task assignment, employee matching, bulk workflows, reports, recommendation sessions and project-management data.
- Built validated CSV, JSON and Parquet ingestion/export pipelines, feature preparation, training-session workflows and inference integration across datasets with 42,052 training rows and 8,924 test rows.
- Integrated Plane REST API and local Qwen/Ollama services; added modular orchestration, reporting, explainability workflows and automated tests for matching, importers and end-to-end pipelines.

UrbanFlow AI – Python Backend & Simulation Engineer

2026

Tech: Python, FastAPI, Uvicorn, Pydantic, httpx, Pandas, SUMO, TraCI, OpenStreetMap, JSON/JSONL

- Designed a FastAPI backend coordinating OpenStreetMap import and normalization, SUMO scenario generation, simulation sessions, road-editor patches, training jobs, metrics and model artifacts.
- Integrated SUMO and TraCI to control traffic lights and simulation entities, including vehicles, pedestrians, public transport and dynamically applied road events.
- Implemented experiment tracking, model registry, JSON checkpoints and generated analytics for simulation runs with 300–500 vehicles and up to 2,241 controlled traffic lights.

Face-Swap-Studio – Python Backend Engineer

2025

Tech: Python, FastAPI, Pydantic, OpenCV, NumPy, InsightFace, ONNX Runtime, file processing, macOS

- Built a FastAPI service for session management, uploads, face detection, visual mapping, image generation, batch processing and result archives.
- Designed a model registry and readiness layer for nine internal and external model backends, separate runners, multi-face processing and per-face assignments.
- Implemented input validation, safe filenames, isolated output directories, temporary-file cleanup, secure downloads and macOS packaging, with optional restoration and upscaling workflows.

Botyan – Async Python Backend Engineer

2023 – 2024

Tech: Python, asyncio, aiogram, discord.py, OpenRouter/OpenAI-compatible APIs, requests/httpx, JSON, Pillow, MoviePy, ffmpeg

- Built an asynchronous Python service integrating Telegram and Discord bots with shared cross-platform conversation history, commands, access control and parallel platform runtimes.
- Implemented JSON-backed persistence for messages, balances, daily rewards, slots, API-key limits and deduplication; supported context windows of up to 100 messages for chat and 1,000 for image-caption workflows.
- Developed backend media-processing workflows with Pillow, MoviePy and ffmpeg for image captions, demotivator generation, Telegram video-note effects and generated media delivery.

Skills

Technical: Python, FastAPI, Pydantic, REST APIs, asyncio, Uvicorn, HTTP clients, Pandas, NumPy, CSV/JSON/Parquet, SQL/SQLite, Docker, Git, OpenCV, ONNX Runtime, SUMO/TraCI, aiogram, discord.py, Pillow, MoviePy, ffmpeg.

Languages: English – C1 (Advanced); Russian – Native.

Education

SAINT PETERSBURG STATE UNIVERSITY OF AEROSPACE INSTRUMENTATION (SUAI) – Saint Petersburg, Russia

- Master's Degree, Management – Strategic Human Resource Management 2027
- Bachelor's Degree, Software Engineering – Software Systems Design 2025
- Professional Retraining Diploma – Web and Multimedia Applications Developer 2024
- Additional Professional Education: 250+ certificates and diplomas from 60+ universities, companies, educational platforms, institutes, and professional training programs worldwide, covering diverse academic and professional fields and disciplines.